

SKILL DEVELOPMENT



Climbing is a complex sport that requires physical, mental, and technical capabilities. Very technical climbs, like *Darkness at Noon*, 5.13a, at Smith Rock, OR, require near-flawless movement, and are ideal for developing climbing skills. ...



Mike Anderson

CHAPTER 3

"Sometimes I'm completely in sync with what the rock can do... some geological quirk has made it into a piece of music which, when you listen to it, makes you dance. And if you really pay attention to the dance, the harder the route gets, the more blank and featureless it is, the better and more complex is the message you get from the rock."
— Johnny Dawes³

INTRODUCTION

When climbers first take up the sport, there is (unfortunately and shockingly) typically very little thought put into *learning to climb*. Consider a typical first climbing experience (almost certainly in a climbing gym these days): The person will learn how to wear a harness, tie-in, and a few belay commands, and then he or she will be unleashed to "climb." These are the minimum skills needed to preserve life, and typically the only instructions provided. Sometimes the novice is accompanied by someone with more experience (but rarely *much* more) who can pass along the standard phrases such as "use your feet," which lack the nuance and context to be of much value to a total novice. Mostly what the "experienced" climber *successfully* passes on is the ultra-competitive, grade-obsessed nature of our sport, encouraging her to attempt increasingly difficult routes until every inch of vertical progress becomes a struggle. This cultural attitude will immediately and regularly hamper the novice's skill development by pressuring her to pass up teachable moments in favor of achieving immediate results, and by discouraging her from attempting to learn more fundamental skills on the most appropriate terrain.

For whatever reason, when novice climbers enter the sport, it is either assumed that they (basically) already know how to climb or the matter isn't considered at all. Perhaps the belief that the act of climbing

is somewhat natural leads to this omission of instruction, just as one wouldn't expect detailed instruction from a track coach on how to run in a circle. From a skill perspective, climbing is infinitely more technically demanding than running—it is much more like basketball or gymnastics. The experience of novice climbers is thus akin to teaching a novice baller, before his first game, how to tie his sneakers but not how to make a layup! The corresponding gymnastics analogy is too dreadful to contemplate....

To make matters worse, the ability to learn new skills fades over time as the climber gains familiarity. The best opportunity to learn occurs when the climber first begins to climb, an opportunity that is often missed. A well-known sports paradox is that most athletes receive the lowest-quality coaching earliest in their career when it is most critical, and only encounter high-caliber coaching later in life (if at all) when it is least needed. (To wit, many of today's NFL quarterbacks rose to that level in part because they were shipped off at an early age to elite quarterback camps, or privately tutored by renowned coaches.)

Applying the paradox to climbing, a sport in which most athletes receive essentially *no* coaching, it's safe to assume that most climbers have a tremendous opportunity to undo numerous bad habits and replace them with good ones. This can be a daunting challenge, but facing it head-on is critical to the goal of continuous improvement. No amount of effort spent on this challenge could be considered "too much," and yet many climbers neglect it. If bad habits were immediately apparent, they could be dealt with at any time, but many do not become limiting for years, and so remain undetected or easily compensated for. Even worse, these are often overarching habits that infiltrate the climber's entire training program, sabotaging every endeavor from gym sessions to megaredpoint projects, stifling development or even causing regression. However, for the climber who seeks to reach his potential, these overarching bad habits must

eventually be replaced with overarching good habits. Before hammering away on the *specific* good habits needed for climbing (such as relaxing the grip or improving footwork), consider the likelihood that more *general* habits such as *how to acquire new skills* need to be relearned and reinforced.

The theories of motor learning and descriptions of fundamental climbing moves have been well-described in other texts (particularly the ground-breaking *Self-Coached Climber*,⁵ which includes a six-chapter presentation of climbing movement and motor learning), so they won't be repeated here. These are important topics, however, and climbers are encouraged to investigate them. Rather than describing particular moves, this chapter will present climbing-specific *principles for effective skill learning* that climbers can apply broadly to skill development. Selected aspects of climbing skill are discussed, focusing on topics that are often poorly understood or regularly cited by climbers as weaknesses (such as footwork, handwork, dynamic moves, and resting). Finally, the integration of skill-development activities into a periodic training program is discussed, along with specific, easy-to-follow exercises for skill practice that enable simultaneous physical and technical progression.

GOOD HABITS FOR EFFICIENT SKILL DEVELOPMENT

The psychologist Anders Ericsson pioneered research into “deliberate practice,” and one of his fundamental discoveries was that the level of mastery one attains with a skill has more to do with how one practices it than merely *how many* repetitions are performed.⁴ The most effective practice obeys these principles:

- Isolate key individual components of a skill
- Perform those components repeatedly
- Incorporate expert feedback to correct deficiencies
- Gradually escalate the difficulty to attain mastery

Therefore, to maximize the benefits of practice time, it is essential to first learn *how to practice*, or more precisely, how to effectively develop and refine climbing skills. These broad principles can be applied to nearly any climbing skill-development activity, and with conscientious effort and repetition will lead to the following good *habits*:



It is essential to set-aside time for *deliberate practice* on appropriate terrain. Unfortunately, many climbers feel the need to only *train* or *perform*, inhibiting their skill development and limiting their climbing potential.

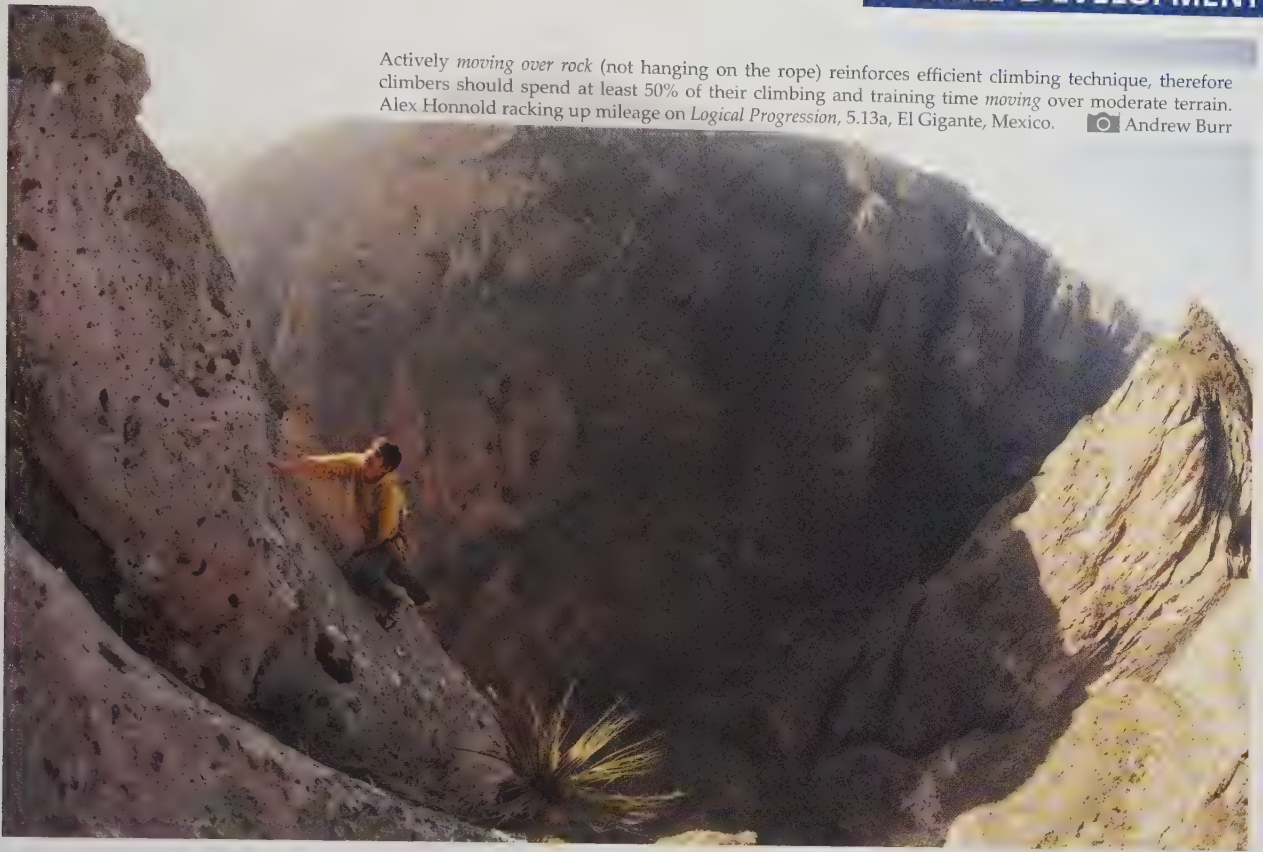
ENGAGE IN PRACTICE

The most fundamental good habit of skill development is also the most neglected: the simple act of *engaging in practice*. Climbing rhetoric itself betrays this fact—while athletes in the ball sports speak of “going to *practice*,” climbers “go climbing” or “*training*.” The difference between training and practice is much more than semantics. Training refers to developing the physical components of the sport: increasing strength, endurance, and flexibility. High levels of

“I wanted to climb as many routes as I could. And the result was that my repertoire of moves increased more and more, and the stamina, too... I gained much more self-confidence. Finally, I started to climb better and better.”

— Josune Bereziartu¹

Actively *moving over rock* (not hanging on the rope) reinforces efficient climbing technique, therefore climbers should spend at least 50% of their climbing and training time *moving* over moderate terrain. Alex Honnold racking up mileage on *Logical Progression*, 5.13a, El Gigante, Mexico.  Andrew Burr



fatigue, intensity, or both are required for effective training. Practice, on the other hand, is a method for gaining skill proficiency through isolation, repetition, feedback, and incremental escalation. It is meditative, requiring concentration, relaxation, moderation, analysis, and a fresh, energetic body.

Practice is critical to every skill-intensive pursuit, including nonathletic pursuits like playing an instrument or speaking a foreign language. While *training* is repetitive at times, the need to optimize physical stimulation necessitates exercises that may not be climbing specific (such as bicep curls). With *practice*

the goal is to optimize neuromuscular stimulation, so the activities used should mimic climbing as much as possible, regardless of the resulting physical stimulation. Additionally, the moves should be worked not to failure, but to perfection.

KEEP YOURSELF MOVING

To foster the best learning, practice should be carried out on appropriate terrain, which leads to the next good habit climbers should embrace: *keep yourself moving*. In climbing, a sport with an infinite number of movement possibilities, practice should consist

Practice and Training Are Not Mutually Exclusive

Novice climbers are often advised to hold off on physical training in order to focus on the higher-priority activities of skill development and movement practice. This advice assumes that the two pursuits are mutually exclusive, but they are not. Through careful planning, it is possible to carve out sufficient time for skill development within a physical training program. Engaging in shorter, but more deliberate skill-development sessions will result in far more effective practice than random climbing activities. Furthermore, most skill development is not physically taxing by design, leaving ample capacity for physical training.

For example, if a typical beginner were to climb every third day in a climbing gym, she could spend the first 30 to 60 minutes on movement drills (which double as a warm-up) and the next 30 to 60 minutes on sport-specific training. Such an approach provides ample time to acquire new skills, and ample opportunity to train in significantly less time than would be consumed by a single day of unfocused cragging. The Rock Prodigy program incorporates skill development in this complementary and synergistic way. To be successful, apply the Good Habits for Efficient Skill Development on a daily basis to make the most of each session.

primarily of simply moving smoothly over mountains of moderate terrain, aka “mileage.” However, climbers often spend too much time on terrain that is too difficult for effective practice, whether bouldering at the gym or projecting at the crag. The very fundamental activity of reinforcement of efficient climbing technique occurs only when the climber is in motion, not when paralyzed by a stopper move or hanging on the end of a rope. Nevertheless, many climbers select routes that are too difficult, resulting in more time spent dangling than actually climbing. There is a time and place for the challenge of a tough climb, but it’s not *every* time and *every* place. One needs look no farther than your typical novice gym climber, who may spend 45 minutes “conquering” a too-hard 30-foot route. On much easier terrain, at a humble pace of ten feet per minute, that novice might have covered nearly 500 feet, which would certainly go much further toward developing and reinforcing efficient climbing technique than struggling up one 30-foot route.

Each climber should consider what percentage of her climbing-related time (whether in the gym, at the crag, or in non-climbing-specific training exercises) is spent *moving over rock* versus doing anything else (resting, strategizing, etc.). Then, she should develop the habit of climbing more on moderate terrain, racking up mileage, and developing and reinforcing efficient movement skills. A good target is to spend more than 50 percent of your total “booted-up” climbing and training time *moving* (don’t include belaying time or long rests between burns). This is especially critical for less-experienced climbers, who will benefit the most from developing fundamental, generic movement skills, rather than dissecting specific crux moves or rehearsing a particular dyno.

Not all climbing time must be set aside for practice, but the trouble is, most climbers set aside *no* time for practice. For many climbers, a simple change in attitude would yield more practice time. Warm-up and cool-down routes are ideal for practice, but only if the climber applies concentration and focus to develop specific movement skills rather than engaging in small talk with the belayer.

LEARN NOVEL SKILLS IN A STRESS-FREE ENVIRONMENT

The general habit of accruing mileage on moderate terrain extends beyond developing efficiency. It is also a prerequisite for learning new moves, and leads to the next good habit: *learn novel skills in a stress-free environment*. Initial learning of any complex skill is inhibited by external stress. Therefore, initial experi-

mentation with novel moves should take place when the climber is relatively fresh, on moderate terrain, with a top rope (or close to the ground, if bouldering), to remove the stress associated with falling. Many climbers may also learn better in private, or in the company of a single supportive companion or coach. Large crowds of onlookers will make some climbers self-conscious, reducing their willingness to attempt fundamental moves or ask basic questions. The ego’s desire to perform in front of an audience is not conducive to this initial learning. As the climber progresses, external stresses should be incrementally increased to make the skill more challenging to execute. This *stress-proofing* process is described in detail at the end of this chapter.

SELECT APPROPRIATE PROJECTS

The habit of keeping yourself moving also applies to the goal routes and boulder problems climbers pursue. This is the next good habit of skill development: *select appropriate projects*. The start-to-finish process of attempting a difficult route and seeing it through to a redpoint is a tremendously valuable learning experience for a climber, so it should occur as often as possible. Unfortunately, climbers often select projects that are slightly too hard and consume one or more entire climbing seasons. A good habit is to spend the majority of the peak climbing season working on routes that can be successfully redpointed in about four climbing days. Such routes are difficult enough to present the climber with the familiar challenges of deciphering difficult moves, dealing with fatigue throughout the climb, planning and executing the projecting process, and coping with the mental stress of sending. Completing four or more such projects in a single season will have an additive effect on the learning and confidence of the climber, while a single, too-hard mega-project that consumes an entire season or more will, at best, yield only one such learning experience, and at worst can be detrimental to the climber’s confidence. Mega-projects have their place (when a “dream route” is within reach, for example), and they offer unique learning opportunities, but from a movement/skill-acquisition perspective, mega-projects offer diminishing returns as the process unfolds. Usually after four or five days the moves are well-known and success hinges on other factors, so make it a habit to select shorter projects as the rule, not the exception.

“The essence of climbing technique hinges on having a keen sense of body awareness. Experience plays a critical role, but the focus of attention is paramount.” — Lynn Hill⁸

PAY ATTENTION

Simply accruing mileage on moderate terrain will improve many climbers’ technique, but that time investment can yield much more improvement if the climber learns to *pay attention* while doing so. Effective practice is more than just robotically going through the motions. Be it spontaneous climbing or specific movement drills, there are subtleties involved in every climbing movement. These subtleties are not always natural, so the body must be consciously directed to perform them while the mind meticulously evaluates the execution and provides feedback. Some moves should be broken down into parts to isolate and rehearse key subtleties. It is only through controlled, escalating repetition with feedback (practice) that these subtleties become habitual, and executed without conscious effort.

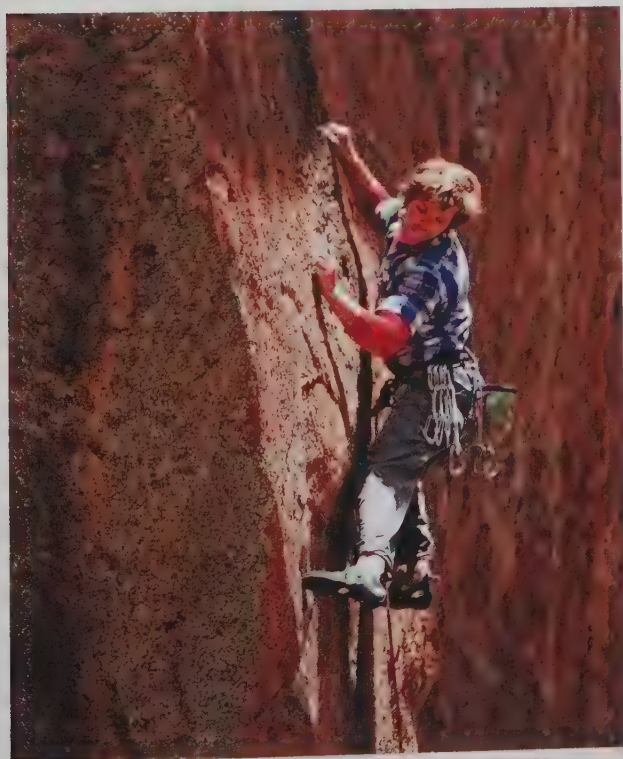
Too often climbers perform movement drills without adequately focusing on the correct details. For example, the ubiquitous “silent feet” drill has the

climber place his feet on the holds silently. The purpose of the drill is to teach accurate foot placement on holds, but the metric for successful accomplishment (sound) is not the best evaluation of this learning objective, as a silent placement could be way off target, or a loud jab could be a bull’s-eye. Therefore, the climber can only expect to learn the desired skill (accurate foot control) if he focuses not just on being quiet, but also on being accurate, and evaluates his performance based on accuracy rather than noise.

Every drill has one or more specific purposes that the practicing climber should understand and pay attention to as the drill is performed. Even other training activities, be they mileage climbing, warming up, endurance workouts, or bouldering sessions, can be greatly enhanced by focusing on a few key learning objectives: skills such as proper breathing, initiating movement in the lower body, etc. (many more will be listed below). High-volume climbing on moderate terrain is a brute-force way to develop more-efficient climbing technique, but with an attentive, meditative mindset, it becomes a scalpel; it becomes *practice*. So pay attention!

FREQUENTLY REPEAT NOVEL AND DIFFICULT SEQUENCES

With practice being the best-known process for developing new skills, and that process consisting of rehearsing novel skills until they are mastered, it is a sad irony that the most novel skills climbers apply in their athletic careers are often never repeated. These are the limit climbing routes and boulder problems that drive our obsession, and oddly, climbers tend to work like hell to make that redpoint but then never climb it again. This is understandable for a faraway route, but this often occurs with local and indoor boulder problems as well. While one ascent indicates some level of proficiency with a given problem, it rarely equals mastery of it; therefore, climbers should make it a habit to *frequently repeat novel and difficult sequences*. This approach is a given in gymnastics. A gymnast would



Matt Segal paying attention to his footwork on *Stingray*, 5.12c, Henry Mountains, UT.  Brendan Nicholson

Good Habits for Skill Development

- Engage in practice
- Keep yourself moving
 - Learn novel skills in a stress-free environment
 - Select appropriate projects
- Pay attention
- Frequently repeat novel and difficult sequences
- Practice drills that target specific weaknesses

not move on from a routine after the first time she sketched her way through it; instead she rehearses it beyond the point of perfection, to the point that it can be performed perfectly *in competition*, under tremendous stress and scrutiny. It is appropriate, then, that the legendary “father of modern bouldering” and former gymnast John Gill was known for repeating and refining his boulder problems until he could execute them in perfect control, without a hint of desperation. In training especially, hard problems that are successfully climbed should become part of the climber’s routine, and oft repeated. Pay attention to the subtleties of the route’s movement, trying to identify the key elements that make each move easier or harder. Refine these subtleties by emphasizing them with each attempt. Experiment with variations in subtle movements, body positions, or alternate sequences, and try to apply novel techniques to the problem. With such focused practice, a once-limit boulder problem

may become routine, and that once-awkward move will assimilate into the climber’s repertoire. This process of “learning-down” a once-desperate sequence will transmit the critical lesson that well-executed movement does indeed make climbing “easier.”

PRACTICE DRILLS THAT TARGET SPECIFIC WEAKNESSES

The final good habit for effectively developing movement skills is to *practice drills that target specific weaknesses*. We addressed identifying strengths and weaknesses in detail in [Chapter Two: Goal-Setting and Planning](#), while the tools for eliminating technical weaknesses follow in this chapter. Just as physical training is planned out in advance to address specific physical weaknesses (such as finger strength), so too does it go with skill development. Recognizing a weakness is the hardest part, but it may be equally difficult

Observation and Imitation

Every climber, novice or expert, should seek to constantly increase his repertoire of climbing moves. Unfortunately, there is no comprehensive “library of climbing movements” to consult, so uncovering new moves can be challenging. Observing other climbers is a great way to identify new movements to add to your quiver. Here are some methods for tapping into your “inner stalker” to augment your movement skills:

Observe other climbers. This is most easily done at a climbing gym, with good lighting and a high density of climbers. For the best results, go to the gym on your rest day, when you can give 100 percent attention to observing others. Identify the more skilled climbers and note how they move, but also pay attention to others. Often a less powerful climber will discover a more efficient sequence out of necessity. Rarely does the strongest climber have the best technique. Take note of new techniques and the problems or routes on which they were applied, and then attempt these climbs on your next gym visit.

Observe elite climbers on video. The proliferation of high-quality climbing films and online videos makes it easy to watch your heroes in action and analyze their movement. Identify novel moves and make a note to try them in your next practice session.

Use video feedback. Any new skill-acquisition process relies on good feedback. Ideally, this can be achieved with a dedicated training partner, but if no such partner is available, a video camera and a cheap tripod can suffice. Film yourself on technically challenging terrain, and then analyze your performance on the next rest day. Are you committing to the most efficient technical solution, or are you attempting to compensate with extra power? This approach works best for stress-proofing skills that are already in your quiver, though not yet perfected. However, sometimes the third-person perspective will allow you to “see” a sequence that was not evident while on the sharp end.

Get the beta. In this Internet age, it seems every route, boulder problem, or plastic gym sequence ever climbed is on video somewhere. Attempt your goal project a few times first (without peeking online), and then review any existing beta (either on video or by watching or talking to other suitors) and note the differences between your sequence and those of others. Are there moves you hadn’t considered? If you find yourself avoiding a particular type of move that seems obvious to others, consider the reasons. Are you masking a technical weakness that needs further attention?

Learn from the masters. If all else fails, it never hurts to take a trip to the zoo and observe the monkeys. These magnificent animals have been climbing 16 hours a day for their entire lives, and they really know how to move. Note how rarely they bend their elbows, how they utilize momentum constantly, and how they plan movements far in advance. Also, note that monkeys *never* use a crimp grip.

to design an appropriate drill to address it. Numerous drills are suggested below, but climbers may need to develop their own. Just as in physical training, devise drills that isolate the key components of the skill and provide feedback for proper execution. These drills should then be performed repeatedly and progressively, as described in the *Progressive Practice* section at the end of the chapter. In general, use the habits discussed above when implementing specific drills.

FOOTWORK

In climbing, mastery of footwork has numerous benefits. For example, foot precision enables the climber to place his or her toes on the best features of a foothold, allowing the feet to support more of the climber's weight (thus reducing the loads and fatigue placed on the upper body). Footwork is often cited as the most important technique to develop, and most climbers consider it a weakness, yet how to develop good footwork remains a mystery to many. Climbers are not born with this weakness; it arises over time from neglect as other skills develop and soon out-

pace footwork. To a large extent, climbers aren't to blame for this—the nature of climbing and the training environment facilitate this neglect. First, the fact that climbing proceeds upward and the hands lead this motion predisposes climbers to focus primarily on their hands while neglecting what the feet are doing. Second, the size and shape of artificial climbing holds are a poor representation of real rock, providing poor feedback to climbers about the quality of their footwork. These factors pervade climbing, so overcoming them requires deliberate action and persistence.

OVERCOMING THE HAND BIAS

PRACTICE DOWNCLIMBING

Moving upward, as climbers do, means the hands lead the way, guided closely by the eyes, which are nearby. This naturally results in tunnel vision, with an overemphasis on hand movements to the exclusion of the rest of the body, specifically the hips, legs, and feet. A straightforward approach to breaking this habit is to practice downclimbing more often. The feet and legs lead the way when downclimbing, requiring the climber to deliberately select footholds, guide her toes to the holds, and apply weight to the footholds. Downclimbing can easily be incorporated into existing training by descending from boulder problems, downclimbing warm-up and cool-down routes, and within endurance workouts (described in [Chapter Five: Base Fitness](#) and [Chapter Eight: Power-Endurance](#)).



Most climbers understand footwork is a critical skill, yet are unsure how to develop it. Fred Gomez demonstrates a few effective techniques for developing footwork on *Rude Boys*, 5.13c, Smith Rock, OR.

Mark Anderson



Down-climbing forces the climber to pay attention to their feet, and is a great way to practice footwork.

“When I first started climbing, my focus was polarized on clutching the handholds and I didn’t pay much attention to how I placed my feet or hands. Over time, I began to understand the finer aspects of how to apply force on a given-shaped hold.” — Lynn Hill⁸

EMPHASIZE FOOT SEQUENCES

This exclusive focus on the hands applies to sequence problem-solving as well, in which climbers routinely neglect the contribution of their feet to the solution. This under-analysis leaves climbers with nothing better to do than to simply use the *largest* footholds nearby, lacking any better ideas. Often, the best foothold is the hold that is in the correct *location*, not simply the largest. The correct location is that which permits the optimal body positions and movement from one position to the next. To overcome this, climbers must make a conscious effort to contemplate the optimal foot positions, which may not coincide with any foothold — a smear may be the best choice.

COMPENSATING FOR CLIMBING GYMS

A wide-ranging root cause of poor footwork among modern climbers is the nature of modern climbing gyms. In particular, route-setting and the size and shape of artificial climbing holds encourage bad footwork habits. In the gym, sloppy footwork is not penalized like it is outdoors, so climbers miss out on this critical feedback. Since climbers spend so much time climbing on manmade walls, it is easy to develop the habit of sloppy footwork. The following practices will help reverse this habit:

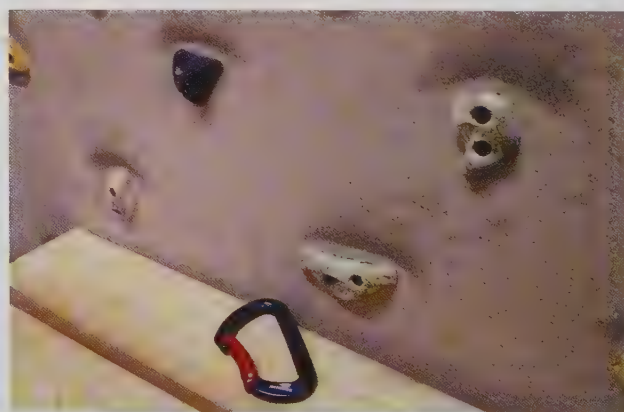
VARY ROUTE-SETTING “RULES” FOR FOOTHOLD USE

The way footholds are incorporated by human route-setters can also bias footwork practice, so climbers must be cognizant of these biases and compensate for them. An “open feet” policy does not force the use of specific holds; therefore, it would not be appropriate for developing proficiency with poor footholds. On the other hand, such a policy is appropriate for developing the feel for optimal foot location because it gives the climber more choices with which to experiment. A “tracking only” policy, in which the hands and feet are restricted to specifically designated holds (usually designated with only the hand movements in mind), tends to force large foot movements that are

not often reflective of rock climbs. Furthermore, the handholds used are typically much larger than most footholds found on rock. Therefore, unless the setter takes extra care to add numerous appropriately sized footholds, the “tracking only” policy has little application to rock climbing and will not improve footwork. Smearing anywhere on the wall should always be “fair game.” Ideally, each climber would use a variety of these methods to ensure exposure to a variety of practice scenarios. If this is not the case at a given gym, bend the “rules” as necessary to provide the appropriate variety of footwork-improvement opportunities.

PRACTICE ON TINY FOOTHOLDS

Compared to natural footholds, artificial holds are much larger, higher profile, and demand very little precision to mount and minimal body tension to maintain. Therefore, *always practice climbing on tiny footholds*. This policy is easy to follow outdoors, so it mostly applies to gym climbing. Climbers should ignore the demands of the route-setter and be their own master, restricting themselves to the smallest foot “chips” and/or natural features embedded in the wall. If these are not available, sub-features of larger holds may be used, such as a smaller edge or ripple, or even the bolt hole. This policy can and should be applied to all training activities, from low-intensity mileage sessions to endurance training to bouldering sessions. For those with home walls, invest in buckets of screw-on footholds and spray them all over the wall. Vary the orientation of the holds so that the best surface is not always available to the climber.



Climbing gyms are great for training, but can be less than ideal for skill development, especially footwork. In the gym, limit yourself to tiny footholds, such as these screw-on “chips”, or “jibs”

WEAR GOOD SHOES

Another simple policy that will effortlessly improve footwork is to *wear good, tight-fitting climbing shoes* when practicing footwork (i.e., *always*). For their non-performance climbing days, many climbers use “training shoes” like an old, blown-out or resoled pair of shoes that are more comfortable. Such shoes are also sloppier, so they do not provide precise feedback to the climber on the quality of foot placements. Furthermore, these old, comfy shoes don’t penalize the climber for clumsy foot placement the way tight-fitting shoes do (like when a climber over-aggressively smacks his toe into the wall — ouch!). For some, the added expense of good-quality shoes may be enough to promote more care with foot placement.

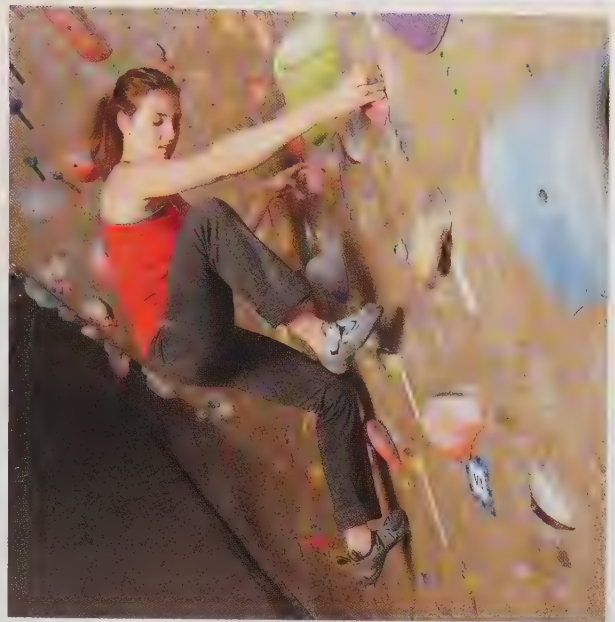
Overall, artificial climbing gyms are a tremendous asset to climbers and are largely responsible for the gains in climbing performance seen over the last two decades. However, a few peculiarities have caused a corresponding drop-off in footwork. The three policies detailed above (vary route setting “rules” for foothold use, practice on tiny footholds, and wear good shoes) are fairly painless, transparent steps climbers can take to replace the sloppy footwork habits inadvertently encouraged by artificial gyms.

FOOT-PLACEMENT GUIDELINES

In addition to the general policies listed above, several more deliberate steps are available to improve the climber’s foot-placement precision and get the most out of footholds. These drills require concentration to perform correctly before they become habitual, so they should be deliberately practiced during targeted sessions set aside for just this purpose, when the climber is not distracted by other objectives.

MAKE MANY FOOT MOVEMENTS

While not a hard-and-fast rule, in general, the feet should move between holds more frequently than the hands. That is to say, for every hand movement, a technically skilled climber will make two or three foot movements. The reason for this is that large foot movements, or highsteps, are generally less precise and require that the climber push his center of mass (COM) out away from the wall to reach the higher foothold. As this happens, the load increases on the climber’s upper body while the direction of pull on the holds rotates outward, reducing the mechanical advantage and further straining the fingers. In addition, the resulting highstep position makes it difficult to fully weight the foot and prevents the climber from



The protruding, “pinch-able” holds typical of climbing gyms encourage the bad habit of over-high-stepping. Instead, make many smaller foot movements when you are practicing in the gym.

initiating the next move in the lower body, which is most efficient.

On the other hand, small foot movements keep the climber’s COM close to the rock throughout the movement, maintaining the optimal direction of pull on the handholds. Furthermore, with the feet on lower holds, it is easier to apply more weight to them, and most efficiently initiate the next moves from the lower body. Gym routes encourage the bad habit of overly highstepping because they tend to lack intermediate footholds, they use protruding handholds that can often be pinched (allowing a more outward direction of pull), and the large footholds demand less precision to mount. This makes highsteps easier in a gym than they would be on the rock, encouraging their overuse. Therefore, during practice, seek to make approximately three foot movements for every one hand movement (actual outdoor routes may dictate a different ratio), even if this requires smearing, and resist the urge to pinch every hold in the gym.

MONITOR FOOT PLACEMENT

Another good foot-placement practice is to “pick a target and watch it land.” This means the climber should deliberately select not just the foothold, but also the precise desired toe location (target), then maintain eye contact with the hold as the toe is placed. It is common for climbers of all abilities to make foot movements without visually verifying the placement, relying instead on *proprioceptive* feedback (spatial

Upward progress on slab climbs relies mostly on friction between the shoe and rock. Weighting the feet and initiating moves in the lower body are mandatory skills. Tommy Caldwell waltzes up *Dancing in the Light*, 5.11b, Squamish, B.C.  Andrew Burr



awareness and touch) to hit the right spot. However, making the most out of small, technical footholds can require millimeter-level accuracy, which is extremely difficult to achieve proprioceptively on the first try. Therefore, when placing the feet, especially on small, technical holds, closely watch the movement as it occurs, and don't look away until the toe has hit the target (a well-positioned partner can provide the best feedback). This process can be very easily drilled during all types of climbing activities.

BLINKING DRILL

Placing the feet on holds proprioceptively may be required in some situations—specifically on steep, strenuous terrain that must be climbed quickly, and when large, forgiving footholds are available. Therefore, it is also helpful to practice proprioceptive foot placement with the *Blinking Drill*, performed by spotting the foothold with eyes open, then closing the eyes as the foot is placed on the hold, and so on. Though the overall purpose is to allow the climber to move faster, it is not necessary or desirable to climb fast when performing the drill. Instead, the climber should select the foothold, carefully memorize its spatial location before blinking, and then carefully move to the hold. He should then observe the tactile feedback from his

toes to assess the placement quality and make necessary adjustments. Once the placement is “finalized,” assess the quality of the placement with open eyes. With practice, the climber should be able to perform this operation more quickly, enabling faster climbing when it counts.

WEIGHT FOOTHOLDS

Precise foot placement is just the first step in properly utilizing the feet, yet many climbers focus on other things once their toes have landed (and sometimes sooner). Most footholds rely on friction between the shoe and the rock to maintain the foot placement, thus shoe manufacturers' closely guarded rubber formulas. The friction force that supports the foot depends not only on the properties of the rubber, but also on the pressure applied to the rock through the foot (called the *normal force*). Therefore, it is critical to develop the habit of pushing the foot into the hold and maintaining that pressure as consistently as possible throughout the move. This can require considerable core strength in some situations, so regular practice is essential. At the point of contact between rubber and rock, increase the mating surface by “wrapping” the toes around the hold as much as possible. Then, strive to keep the surfaces mated to each other with-

Habits of Good Footwork

Overcoming the Hand Bias

- Practice downclimbing
- Emphasize foot sequences

Compensating for Climbing Gyms

- Vary route-setting “rules” for foothold use
- Practice on tiny footholds
- Wear good shoes

Foot-Placement Guidelines

- Make many foot movements
- Select the target and watch the toe land—monitor foot placement
- Use the Blinking Drill to practice autonomous foot placement
- Maintain gradual, consistent pressure on the foothold throughout its use
- Seek out and master technical face and slab climbs

out slippage or rotation. This demands flexing at the ankle and forefoot as the body moves up on the hold. Throughout the movement, concentrate on maintaining a constant force on the foothold, as overloading or underloading it will alter the friction-force interaction, possibly leading to a foot slip.

TECHNICAL FACE AND SLAB CLIMBING

Finally, a great step climbers can take to hone their footwork is to seek out and master technical face and slab climbs. These routes *demand* excellent footwork, and thus provide expert feedback to the climber. As a result, the climber can't help but to practice precise footwork, or walk away in disgust. Slab climbs (routes that are less than vertical) should be sufficiently difficult that they rely on very small friction holds. These demand very precise foot placements, and will not tolerate large variations in the magnitude and direction of the force applied by the foot as the climber moves past the hold. They also force the climber to put weight on his feet and initiate movement in the lower body. Thus, on slab climbs the climber is required to pay attention to these details and reinforce them as habits.

Technical face climbs are those in the realm of vertical, with tiny holds, especially pockets. These routes require the climber to hold her center of mass close to the rock in order to get purchase on the tiny handholds, forcing many small foot movements, with limited ability to lean out and spy holds. The numerous foot movements force the climber to carefully weight and unweight each hold, shifting balance around as

she inches up the face, thus providing ample repetition of these delicate processes. Pockets and other small footholds demand very precise placement and careful application of pressure to remain engaged on the hold. The toughest face cruxes will demand the careful distribution of weight across the hand- and footholds, teaching the climber to sense how much each hold can bear, and training her to apply the correct load. These critical skills cannot be obtained any other way.



The tiny, insecure holds common to vertical face climbs demand precision and force the climber to stay close to the rock, preventing large high-steps. Tara Brouwer hugging *Sister of Mercy*, 5.12b, Penitente Canyon, CO.
 © Mike Anderson

“I concentrate especially hard on my hands. I take time to feel the holds, not only to find the best way to grab them, but also to find out what is the best way to put my feet on them later.” — François Legrand⁷

HANDWORK

Few would argue that the connection between the hands and the rock is critical to difficult climbing, so learning how to get the most from this interface is fundamental to improving a climber’s technical skills. Every climber should have a basic working knowledge of the various hand positions, including but not limited to: the open grip, crimp grip (including the closed crimp and semi-closed crimp), pinch, sidepull, gaston, false grip, mantel, and undercling. If unfamiliar with these grip positions, refer to Craig Luebben’s *Rock Climbing: Mastering Basic Skills* or similar texts.⁶

There are an infinite variety of handholds, and few are exactly alike. Although training tools and gym holds are generally ergonomically shaped with smooth contours, real rock rarely is. Handholds are often highly irregular, requiring precise placement of each individual finger to gain the most purchase on the hold. In some cases, fingers need to be applied in a particular sequence for maximum benefit.

Determining the most effective way to grasp a hold requires a combination of visual and tactile feedback. Various factors come into play depending on the hold type, but with practice, an experienced climber will be able to regularly select the proper hand configuration on sight. This section will provide a variety of options to consider when determining how best to grasp a hold in the real world. Experiment with these suggestions on the rock to get the most out of each handhold.

EDGES

Most edges are not perfectly flat, are not parallel with the ground, and vary in depth. Generally the most important factor is depth, so the primary aim should be to get the two strongest fingers (the middle and ring) on the deepest part of the hold. Next consider the angle of pull, which is usually the direction the forearm is pointing. All else being equal, the part of the edge perpendicular to the angle of pull will provide the best grip. For irregularly shaped edges, split grips or finger stacks may be useful to get the most out of the hold.

Consider different hand positions such as a closed crimp, semi-closed crimp, or open grip. Often the geometry of the hold will dictate a particular hand position. Perfectly flat holds will require a closed or semi-closed crimp, whereas one may benefit from the use of an open grip on certain convex, arcing edges. Often by initially grasping an edge in an open position and then “rolling it up” into a closed crimp, it’s possible to maximize skin contact to get a smidgen of extra purchase. Realize, however, that this is hellish on the skin and can cause tendon and ligament injury when performed under heavy load.

POCKETS

Pockets are almost always positive in one direction, but the majority of the time that direction is not straight down. As a result, most pockets will provide a better grip when used as a sidepull, gaston, or undercling. In some cases it may make sense to use a single pocket in two or three different positions as you move past it. Poke around to determine the depth of the pocket, and note any unusual features inside the cavity or on the lip that might provide extra grip. Test the pocket with every finger (or combination of fingers) to determine which finger or set of fingers provides the best grip (for example, for a two-finger pocket, consider index-middle, middle-ring, and ring-pinkie). Often deep pockets will feel better if the fingers are *not* stuffed all the way in. For highly irregular pockets, it may be necessary to insert fingers sequentially to get the best grip.

Occasionally, the position of the inactive fingers can affect how secure or “tweaky” a given pocket feels. For two-finger pockets that use index-middle, it usually feels best to curl the ring and pinkie fingers down into a fist. However, some shallow middle-ring two-finger pockets feel better with the pinkie extended. It may be possible to scum the inactive fingers on knobs or rough patches adjacent to the pocket. In extreme situations, it may make sense to wrap or stack a free finger over one of the fingers in the pocket (as for a closed crimp), but be wary of injury when attempting this.

Finally, pockets present unique injury risks that are worth discussing. First, pockets should almost always be grasped with an open grip, except where it’s possible to fit three or more fingers into the pocket. Crimp positions place severe amounts of stress on the *proximal interphalangeal* (PIP — the big knuckle) joints of the fingers used, so climbers shouldn’t use these grips unless they have adequately trained them. As with fingerlocks, the collateral ligaments of the

Pockets of Subtlety

On the milky-white limestone in the backwoods of the Rockies is a magnificent pocketed testpiece. The “business” starts immediately with a brutal V9 boulder problem that leads into a 40-foot face-climbing gauntlet of tweeky, tenuous sequences, culminating in the redpoint crux: a long windmill move off a sloping two-finger pocket. On two trips spread over a year, I fell on this move repeatedly on redpoint as I slipped out of that pocket mid-windmill. My frustration was compounded by my ability to nail the move every time off the hang. One burn, while groping that pocket and experimenting with different grip positions, I noticed that if I rotated the grip clockwise slightly, toward a sidepull, the friction was just a smidgen better, the pocket’s geometry made the grip just a little more positive, and I was able to get an ever-so-slight dime-edge thumb catch. These miniscule differences were imperceptible to the eye, but the fingers knew.... When you are on a route right at your limit, little details like this can be the tipping point that makes or breaks a send. That was the case this time, and I sent the route on my next go.

— Mike Anderson

finger joints are at risk during big moves off “locker” pockets. Take care when moving past pockets to limit the torsional (twisting) forces on these joints. Ideally, the fingers will have enough room to rotate freely in the pocket as the angle of pull changes.

SLOPERS

Two factors generally determine the best way to grasp a sloper: the friction and steepness of the contact sur-

face. The first order of business is to locate the most positive point on the hold. This is almost always the best place to grasp the sloper, unless it is more slippery than another area that is only slightly steeper (which may be the case on well-traveled routes with polished handholds). Experiment with different areas to determine which spot provides the most secure grip; usually, feeling around for the various options with your hand is the best way to determine how to



best grip a sloper. Often large slopers will have irregularities on their surface that can be exploited for additional staying power. Align these irregularities with a finger-pad or skin ripple.

PINCHES

Most pinches vary in depth and width to some degree. Pinch strength typically varies depending on the width of the pinch involved, so experiment with different grip points. Look for irregularities in the surface of the hold, where extra friction can be utilized, or areas that are more positive, and thus allow for more outward pull. Occasionally, altering body positions can transform a desperate pinch hold into a more favorable sidepull, sloper, or gaston, so consider the alternatives to pinching.

THUMBS

After the head, thumbs are perhaps the appendage most underutilized by climbers. Nearly every hold offers a thumb catch (not that a thumb catch should always be used, but it should at least be considered). Thumbs can also wrap over the side or corner of an edge, or protruding knobs near edges, pockets, or slopers. Thumbs can be used to gaston vertical edges or pockets that are too distant for a traditional gaston, or to undercling ("gasto-cling") distant overlaps.

Finally, as discussed in the footwork section, most modern gyms are not intended to realistically represent outdoor climbing. Gyms have a disproportionate amount of slopers and pinches, so be mindful of the type of holds chosen to train and practice on. Ideally, these holds would mimic the type of holds found on performance objectives. A few hold manufacturers still make irregularly shaped plastic holds, so consider finding such grips, to practice quick identification and selection of ideal hand positions. If all this hand-placement discussion sounds like nonsense, consider that elite competition climbers memorize artificial holds from the top manufacturers in order to quickly identify them from the ground, thus ensuring optimal hand placement upon initial contact. Paying attention to every weakness the rock has to offer and learning how best to exploit them can spell the difference between success and failure.

Tips for Better Handwork

- Experiment with various grip positions to find the right fit
- Use the deepest and/or most positive aspects of edges and pinches
- Consider injury risks when evaluating the best use of pockets
- Slopers are rarely uniform; identify the spots that are more positive or offer better friction
- Consider using pinches as sidepulls or gastons
- Get the thumbs involved with catches, wraps, and distant "gasto-clings"



Thumbs can be employed to exploit distant features, such as to gaston vertical edges and pockets, or to undercling distant overlaps. Zak Roper sticking a dynamic "thumb-er-cling" on his project at the New River Gorge. © Dan Brayack

DYNAMIC MOVEMENT

Other texts such as *The Self-Coached Climber* go into extensive detail on the topic of dynamic movement, so this discussion is limited to practical approaches for incorporating dynamic-movement practice into a periodic training routine. Mastering dynamic movement has tremendous advantages. Many climbers think dynos only apply to the most difficult crux moves, but with improved skills, dynos can have much broader application. Consider Basque climbing legend Josune Bereziartu (the first woman to climb 5.14b, 5.14c, and 5.14d), who has perfected her movement and confidence to the point that virtually every movement she makes is a deadpoint, thus sparing tremendous

Girl Power

I remember watching my husband, Mike, effortlessly float up and down the campus board, and I was quite sure that I would never be able to do such magic. Nevertheless, Mike insisted that I try it, and I relented. I remember grabbing the campus rungs the first time and staring straight up. What had I agreed to? I felt like a ton of bricks, and I struggled to let go of that first rung as my mind fought to maintain control. Mike took off some weight with a power spot, just enough that I could bump my right hand to the next rung. Then my left, followed by the right, and so on—all the way up the ladder I went. It was awesome! I was pretty sure Mike carried me to the top, but he insisted that he only helped a little. Campusing was possible; now it was on!

With each attempt, I grew stronger and more confident. I required less assistance each workout and made tangible progress both in training and on the rock. My movements became more precise and I gained the ability to “explode.” Soon I was able to campus alone. More importantly, my climbing changed. My confidence skyrocketed, I could deadpoint much better, and my new “go for it” attitude made me more decisive and explosive. I used to waste a lot of time on routes hesitating, wasting precious energy. This hesitation allowed my mind to wander to negative self-talk, which was very harmful to my climbing.

Learning to make quick and decisive movements on the campus board brought me a new focus in climbing. Finally, I had unlocked new levels of confidence! A whole new world opened up, and climbing became much more fun.

— Janelle Anderson

amounts of energy throughout the course of a long ascent.

For many, climbing dynamically is a psychological challenge motivated by the reluctance to relinquish control on the rock. This manifests in a fear of letting go and a desire to remain static in every movement. The campus board (discussed in detail in [Chapter Seven: Power](#)) is arguably the best tonic for this condition, as dynoing is nearly impossible to avoid on a campus board. As with all skill development, once a weakness has been identified, practice the new skill in a consequence-free environment, and gradually stress-proof the new technique. Establishing a safe landing zone is the first step to executing any dyno, so take the time to ensure a good distribution of crashpads, spotters, or a safe belay. Using a *power spot* (where the spotter assists the climber through a difficult move by pushing gently around the lower back) can be extremely helpful for timid dynamic climbers.



Many climbers struggle on dynamic moves because of a reluctance to relinquish control of their body. The Campus board is a great practice tool for overcoming such hesitation.

When executing difficult dynos, consider the role of the trailing hand as well as the moving hand. Aaron Parlier checks his swing on *Pocket Rocket*, V6, Joe's Valley, UT. © Dan Brayack



The campus board is also a valuable practice tool for more advanced dynoers, who will find it helpful for perfecting their deadpoint skills. The smooth rungs encourage a “go for it” mentality, and the ability to progressively increase the distance between holds allows climbers to methodically escalate the movement difficulty, making for ideal practice.

Bouldering is also an excellent platform for climbers of all abilities seeking to practice their dynamic skills, provided problems are set in a manner that mandates dynamic movement. Bouldering allows for much more complicated application of dynamic movement, requiring coordination across the entire body in multiple planes and axes of travel.

When executing a difficult dyno, most climbers focus on the moving hand, and the force and movement required by that hand to reach and latch the finishing hold. Yet you should also consider the forces and motion required to generate upward momentum from the lower body before the leading hand releases from the starting hold, consider the role of the trail-

ing hand, and the path of the hips throughout the dyno. Often the key to a difficult, foot-cutting dyno is controlling the swing of the body away from the wall. The trailing hand is the key to controlling this motion, so focus on pressing the lower hand into an effective mantel position as your center of mass rises. Extend the lower arm as the hips move away from the wall to allow consistent pressure on the lower hold, and practice slowly and gradually reducing the outward momentum, thus avoiding jerky decelerations. In addition, the forces applied through the feet (and the timing thereof) in the moments before they cut loose may be critical to controlling the swing, so consider their role as well.

REST TECHNIQUES

Resting is an essential skill, and it's a good idea to experiment with rests before one is desperately needed. In climbing, rests are used to alleviate the load on the forearm flexors allowing them to de-pump. Any body position that permits a relaxed grip can be a rest, and this provides a lot

of possibilities, many of which will be described here. When first arriving at a rest, spend some time exploring the possibilities (various hand positions, hidden holds, thumb catches, scumming other body parts, etc.). Within the first minute or so, the climber should pick the best, most relaxing option and commit to it.



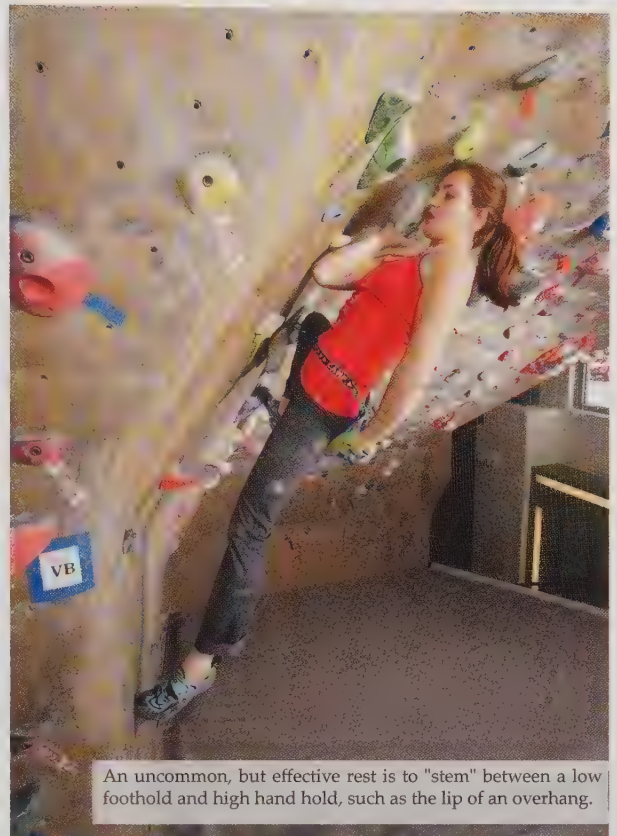
Positive “jug” holds make good rests because they provide a mechanical advantage. The supporting force of the rock is applied closer to the finger joints rather than the finger tips, which shortens the lever arm, and reduces the *moment* on the joint. This translates into a lower force that must be applied by the finger flexor muscles to hang onto the hold.

Continually groping around is less restful, and there is a learning curve that improves a singular resting position with practice; shifting around prevents this beneficial adaptation.

Practicing various rest positions in advance will make for quicker recognition of the optimal positions when climbing, improving the quality of recovery. Remember, the point of resting is to permit a relaxed grip so that the forearms can de-pump, and this does not happen automatically—it must be consciously directed. Once established in the rest, deliberately shift weight to the feet as much as possible, and then gradually loosen the grip of each hand as much as possible, carefully observing for slippage.

A large hold, or jug, is always a good rest. On steep terrain, a jug gives the climber an improved mechanical advantage by allowing him to apply force to the hold closer to the proximal joints of the fingers rather than the fingertips. This requires less force to be applied by the forearm flexors, permitting a relaxed grip. Thus, the best rest will be had by contacting the lip of the hold as close to the forearms as possible, which could mean hanging off the palm of the hand by bending the arm at the wrist. On very positive jugs, the protruding bone at the base of the hand (the *pisi-form*) can be “hooked” over the lip or even slotted be-

On very positive jugs, the protruding bone at the base of the hand (the *pisi-form*) can be “hooked” over the lip, or slotted behind deep flakes.  Mike Anderson



An uncommon, but effective rest is to “stem” between a low foothold and high hand hold, such as the lip of an overhang.

hind deep flakes. This simultaneously holds the body close to the rock and relieves the forearms, providing a great rest. On steep terrain, the climber should hang with straight arms and rotate the hips with a backstep to bring the center of mass closer to the rock, improving the direction of pull. For large holds on less-steep terrain, it may be more restful to mantel onto the hold and *stand* on it (while leaning into the wall), versus hanging off it by the arms.

Body parts can often be wedged into cracks or pockets, providing an excellent rest. Experienced crack climbers know this, and are always on the lookout for hand jams and fingerlocks during difficult sport climbs. For those who aren’t crack climbers, seek instruction from an experienced crack master while experimenting with different jams close to the ground. The nuances of jamming are best learned through personal experimentation, and not during a difficult climb. The portions of the crack that form natural constrictions will offer the best jams, and the climber should seek to rest on his skeleton with a relaxed grip, not the finger-pads. Good jams can be painful, so pain (or lack thereof) is not a good indicator of a jam’s quality. Athletic tape is often used to reduce pain and get the most out of a jam. Foot, knee, arm, and even head jams are possible, so cast a wide net. The time spent

exploring the possibilities will not be in vain.

The legs are critical to any rest, by supporting whatever load is shed by the arms, but the legs and feet can be loaded in multiple unusual ways. Besides standing directly over them, the next most common use for the feet is the heel hook. It is best performed on a large, positive jug, but it doesn't have to be. Generally, the higher the hook, the greater the load it can bear. Heel-toe cams, calf hooks, or sideways slotted heel hooks are even better because these improve the mechanical advantage, allowing more weight transfer to the lower body. This is similar to the way a positive jug is less taxing for the forearms. A double-heel-hook "bat hang" looks impressive, but is often not very restful or worth the effort to get into.

Other leg rests include stems and kneebars. Stems are best used in corner systems, which allow the climber to lean in over the feet. Climbers should try to position their center of mass over the imaginary line formed between the two feet, or as near as possible. Sometimes this can be accomplished by crouch-

ing low and using backsteps or drop-knees to lower the center of mass. During a stem, the feet do not have to be level with each other, nor should stems be limited to the feet. It is often possible to stem with the feet on a low foothold and the hands pushing off the lip of an overhang, or use the heel of one hand opposed to the opposite foot. Here again, the goal is to get the center of mass hovered over the base of support formed between the feet and hand (or shoulder blade or head).

A kneebar is a type of jam utilizing the entire lower leg. Ankle flexion is necessary to adjust the size of the jam and apply the opposing pressure that locks it in place. The quality of the kneebar depends on the geometry of the rock and the climber. Generally, the best kneebars are a tight fit with a positive knee catch and foothold, and require little effort to maintain. Kneebars that don't fit tightly will require significant calf flexion and can be very strenuous. Kneebars with poor footholds or insecure knee catches will require more concentration and physical effort to maintain, and will also be less restful. A sticky-rubber kneepad improves the holding power between the rock and knee, and can improve an otherwise poor kneebar. They also improve the level of comfort.

Finally, a great rest position that is often overlooked is the foot-squat. This is used on less-steep terrain, or in a cave feature with a large foothold. The climber plants one foot on the hold, dangles the other foot lower than the first, and then squats down over the engaged foot as low as possible in an attempt to balance his center of mass over the foothold. In a cave or scoop, a hand can sometimes push off the lip of the cave to lock the climber in place. When this isn't possible, the lower dangling foot can be flagged (pressed into the wall), resisting gravity's pull on the climber away from the rock. With an adequate positive high foothold, this hooking action can provide a no-hands rest.

"I feel that the best way to gain strength in training is to do it on plastic. But you must be aware that you need time to transfer the results of those workouts to natural rock."

— Josune Bereziartu²



The "foot-squat" is an excellent rest that is possible with a protruding foothold or ledge on lower-angled terrain. The climber plants one foot on the protruding hold and squats down on it while flagging the other foot as low as possible to balance her center of mass over the high foot.

Mike Anderson

The Beta Trap

"Bachar reckoned he had got this trait from John Gill: never tell anyone how to do a problem. Let them figure it out, because it's part of the problem." — Jerry Moffat⁹

Discovering the solution to a puzzling sequence can be extremely satisfying, but these days it's very easy to bum the beta from a buddy or find it on YouTube rather than figure it out for yourself. Borrowing beta is most certainly the fastest way to dispatch a project, but it is a crutch that, if used too often, will develop into a significant weakness. It's possible to navigate a climbing career for some time by relying on borrowed beta, but eventually the beta well will dry up. Be it a difficult onsight, a first ascent, or simply a route off the beaten path, at some point every dedicated climber will face a confounding crux with only his wits to solve it. Furthermore, each climber has unique attributes: physical (height, wingspan, ape index, and finger size), psychological, and physiological (strength, power, endurance, flexibility, technique). Therefore, one climber's beta may not suit another. Don't make the crux harder than necessary by forcing a sequence that was developed by another climber.

Deciphering complex sequences is a skill that requires practice. It can also be great fun and extremely satisfying. Take the time to work out the beta on your own (most of the time) in order to ensure that this skill is available when you need it most. Once you *think* you have it worked out, you can still go to YouTube for a second opinion.

As John Bachar said, figuring out the beta is part of the problem—a part of what makes a problem hard, for sure, but also a part of what makes it fun and rewarding. Welcome the opportunity to discover your own solution from time to time, and don't be ashamed to feel that much more proud when you send.



Heel-toe cams, calf hooks, or sideways slotted heel hooks (shown here) make great rests because they transfer more weight to the lower body. Rob Pizem taking it free and easy during the FFA of the Wiggins Route 5.12, UT. Andrew Burr

PROGRESSIVE PRACTICE

For optimum learning of any new skill, the practice activities should be repetitive and progressive, which is to say, the activities increase in difficulty as the proficiency increases. In terms of climbing, this progression can be accomplished by proceeding toward more real-world climbing situations, an environment inherently more challenging than the controlled arena of the initial skill-acquisition setting. In general, a progression toward realistic climbing scenarios can proceed as shown in the accompanying figure, with increasing difficulty arising from one of several sources: the activity type, intensity, climbing situation, audience, or various combinations thereof.

For example, a climber seeking more precise footwork begins with the Precision Feet drill performed as practice during low-intensity bouldering traverses. Over time, his footwork improves during these sessions, but it may not be evident in his climbing yet, so he includes this drill in indoor toproping sessions, both low and medium intensity. As his footwork improves in these domains, he includes the drill during power-endurance training sessions that are performed by bouldering in a gym. He may also include the drill during warm-up or non-limit climbing routes on rock. As he gains further proficiency, he includes the drill in indoor limit (high-intensity) bouldering sessions, or route-lapping endurance-training sessions outdoors.

These transitions from one combination of activities to another provide a logical transition that draws the newly acquired skills deeper into the climber’s repertoire. These are incremental steps that are attainable and enable the climber to build on past success and familiarity with the skill. Over time, this process brings the skill up to a level of proficiency and familiarity where it could actually be used when it counts — on limit climbs on real rock.

To further illustrate this process, the training calendars presented in [Chapter Ten: Building a Training Plan and Other Training Considerations](#) include recommendations for planning practice, progressing from Skill Acquisition, to Skill Practice, and finally to Stress-Proofing. This demonstrates how practice activities can be premeditated and integrated into the physical training plan for a given season to create a progressive skill-development plan. As discussed in [Chapter Two: Goal-Setting and Planning](#), at the beginning of each season, the climber should clearly identify a limited number of skill-development goals and plan how they will be reached, using the calendar as a template. It is best to focus on only a few key goals each season, so that adequate attention can be applied to each skill. Planning to work on “everything” guarantees that the climber will improve at nothing.

MOVEMENT DRILLS

Numerous movement drills are suggested below that may be incorporated into practice sessions. They are categorized according to aspects of climbing skill, and should be selected to address a specific skill or weakness. After identifying a technical weakness, select the appropriate drill. Don’t try to accomplish too many drills in a given practice session—select no more than three emphasis areas for each session, and concentrate on those.

Drills should initially be performed on moderate terrain, with the primary goal of *learning*. Therefore, drilling climbers shouldn’t feel pressure to send, only to learn effectively. Falling, or otherwise failing at a drill is a part of the learning process, and should be common. If possible, seek a quiet time for this practice (a quiet corner of the gym, or off-peak hours), away from distractions. For the best results, drills should be revisited, and as skills improve, the difficulty can be increased progressively as described above in Progressive Practice.

Activity:	Practice	Training	Performance	
Intensity:	Low	Medium	High	
Scenario:	Low Bouldering	Top-roping	Leading	High Bouldering
Audience:	Solo	Coach/Partner	Friends	Public Crowd

Indoor Artificial Arena
Less Climbing-Specific

Practice Escalation

Outdoor Realistic Arena
More Climbing-Specific

Skill development should be progressive, just like physical training, progressing toward realistic outdoor climbing along these four attributes.

General Climbing-Efficiency Drills

Beginner Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Straight Arms	While traversing an indoor circuit, attempt to keep the elbows virtually locked, by hanging with straight arms and a low center of gravity. Utilize flags, backsteps, and crosses to maintain straight arms while traversing across the wall.	Elbow bend. Also, note the amount of hip and leg movement needed to keep arms straight	Smaller holds, faster pace, and/or incorporate more vertical (up or down) movements in the circuit	Focus on adjusting foot and hip positioning to permit longer movements with minimal elbow flexion.
Objective: Reduce reliance on upper body				

Intermediate Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Rest Practice	While traversing an indoor circuit or climbing an outdoor warm-up, seek and practice novel rest stances, especially "no-hands" or other easy stances involving stems, kneebars, drop-knees, jams, thumb wraps, etc.	Quality of forearm recovery, level of relaxation and calm, reductions in breathing and heart rate	Steeper terrain, more difficult terrain, more strenuous rests	Focus on quickly finding the best rest position, and then settling into it and progressively relaxing all muscles not needed for the rest. Slow your breathing and concentrate on relaxing the forearms.
Speed Climbing	While toppling a familiar route indoors or out, climb as fast as you can. Time yourself and try to improve each attempt. Mentally rehearse the route beforehand to speed yourself up. Efficiency is not critical, so go "all out." The goal is to extend your range of climbing pace so that you can climb faster when appropriate.	Ascent time, which can be compared with your partner's	More difficult terrain, and more technical routes	Focus on moving continuously. Don't analyze every move; just make a decision quickly and commit to it. The route should be easy enough that it isn't necessary to nail the sequence perfectly.
Objective: Increase climbing pace				

Advanced Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Finding Calm	While ARCing (boulder or TR), plan a course that incorporates several dynamic moves. Consciously pause after each dyno to restart breathing with slow, deep breaths. Deliberately relax the grip, shift weight onto the feet, and if possible shake out.	Have a partner monitor your breathing and provide feedback	Incorporate with progressively more realistic climbing situations	Focus on your thoughts and feelings during and immediately after dynos or strenuous moves. Try to identify the moment that breathing stops or you tense up. Practice overcoming this response.
Objective: Maintaining calm through strenuous moves				
Movement Perfection	While bouldering indoors or outside, select a problem near your flash level and repeatedly climb the problem over several sessions, attempting to complete it as efficiently/effortlessly as possible. This drill can span seasons or years for some problems.	Partner's observation and personal evaluation of effort. Monitor breathing and check for grunting.	Increase the difficulty of sequences/problems used, increase climbing pace	Focus on deadpointing dynamic moves, grip control, climbing with good "flow," maximizing lower-body contribution, and eliminating points of hesitation.
Objective: Perfect a complex sequence				
Momentum Climbing	While toppling or bouldering on moderate terrain, attempt to climb the route in one continuous movement. Try to minimize any pause in motion. Use a well-known route, at first, and mentally rehearse the sequences before beginning. As you progress, attempt this drill on lesser-known routes.	Partner's observation and personal evaluation of effort required.	Steeper, more difficult terrain. Attempt this drill with higher states of fatigue or on lesser-known routes	Focus on propelling your ascent with your legs, using your arms mostly as "guides," as if you were hiking up a steep slope. Imagine yourself flowing like water or dancing a continuous choreography.
Objective: Develop a flowing, dynamic style				

Lower-Body Involvement Drills

Beginner Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Stuffer Step <u>Objective:</u> Encourage many foot moves and reduce highstepping	While boulder traversing or top roping, the climber will make three foot movements for every one hand movement. Foot moves should be small and should allow the climber to fully weight his feet. If in a climbing gym, avoid pinching the handholds, which makes them unrealistically "positive."	Feeling of weighted feet, and initiating movements from the lower body	Steeper terrain, more difficult terrain, perform faster	Focus on keeping the COM close to the rock throughout the entirety of the movement. Initiate movement in the lower body by pressing with the toes, feet, and legs.
Thumbs Only <u>Objective:</u> Thumb use, and engaging the lower body	While top roping a moderate, just-under-vertical route, attempt to climb using only the thumbs (and feet). Use a well-featured route and make small foot movements. Attempt to maintain smooth, flowing movement, which requires significant effort and control from the lower body. This drill can also be done with "two fingers only."	Smooth movement	Steeper terrain, longer reaches between holds (above the shoulders), faster climbing pace	Focus on adjusting body position to maximize lower body contribution (moving hips side-to-side to distribute weight over footholds). Note how the distribution evolves through each move.

Intermediate Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Downclimbing <u>Objective:</u> Overcome the hand bias	On any terrain, indoors or out, climb down in addition to climbing up. Deliberately select footholds, guide your feet to them, and don't look away until they "land." Balance your body before moving the leading foot, and avoid large foot movements.	Feel the pressure placed on the footholds and how your lower body supports your weight	Steeper terrain, more difficult routes	Focus on maximizing weight on your feet and relaxing your grip accordingly. Your focus on footholds should be increased, but also ensure that you are weighting them fully.

Advanced Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Toeing Hard <u>Objective:</u> Maximizing weight on feet	On moderate terrain, attempt long reaches while concentrating on maximizing foot force throughout. The mind will try to focus exclusively on guiding the reaching hand, but also focus on the opposing foot, which drives the movement. Simultaneously guiding hands and feet in different tasks is difficult but critical.	Feeling of foot pressure through the move	Use longer reaches and dynos, less-positive feet, steeper terrain, and target handholds that require more focus to hit (i.e., pockets)	Focus on maintaining steady force on the footholds throughout the move. As the handhold nears, it will distract you, but stay focused on the feet.
Pulling Feet <u>Objective:</u> Improve efficiency on steep terrain	While top roping on steep terrain with positive footholds, deliberately pull with your feet to keep your COM close to the wall. Make careful toe placements in the incut part of the hold, like fingers for max pulling power. Experiment with pulling and pushing in unison. Try it on a traverse as well, pulling with the leading foot.	Observe the direction of pull and/or push between the hold and foot as you move past each hold	Use less-positive footholds to increase the complexity of required force application	Focus on applying the foot force in the best possible direction (not necessarily straight down) to hold yourself close to the wall. Feel the force applied by the foot and how its direction evolves as you move.



Target Camping is an advanced drill for developing dynamic accuracy. Here the climber has marked out a "target" for his left hand and is camping to it.



The Foot Stab Drill is a straight-forward way to practice precise foot movements. Simply pick a target on the wall and practice touching it with your toe.



The Pulling Feet Drill promotes keeping your COM close to the wall on steep terrain by pulling in on positive footholds.

Footwork Drills

Beginner Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Precision Feet <u>Objective:</u> Precise Footwork	While boulder traversing or top roping, make careful and precise foot placements. Select the bull's-eye on each foothold deliberately, and don't look away until your foot has reached the target. Take your time initially, then speed up as you gain proficiency. Sound provides immediate feedback, but toe accuracy is the ultimate goal, so focus on that.	Sound and feel can indicate precision; have a "referee" ensure you don't look away too soon.	Combine with Blinking Drill, training exercises (PE, Limit Bouldering)	Focus on controlling the movement of the foot toward the hold; bull's-eye on the first shot. Maintain concentration until each foot is placed correctly. Don't think ahead to the next move before completing the current move.
Foot Stab <u>Objective:</u> Improve foot coordination	While standing near any wall with distinguishable features (not while climbing), stand on one leg, then pick out precise "targets" on the wall to aim for and touch with the toe of the free leg. Can be performed statically or dynamically.	Precise placement of toe on target; balance	Smaller targets, more distant, off-balance points. Perform from a hanging position, while climbing, and/or attempt "foot dynos"	Focus on accuracy, maintaining balance throughout the rep, moving decisively, and focusing the eyes on the target prior to movement.

Intermediate Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Blinking Drill <u>Objective:</u> Develop proprioceptive foot control	While bouldering or top roping, visually identify the desired foothold, memorize its spatial location, and then close your eyes while guiding your foot to the hold, making any necessary adjustments by feel. Open your eyes to assess your performance, spot the next hold, and so on. Can also be performed with hand movements.	Toe feel and visual assessment of placement	Perform faster, move the hips simultaneously with foot movements	Focus on visualizing the location of the foothold, how it feels to gain the hold, and assessing the quality of a foothold and foot placement by feel.
Jibs Only <u>Objective:</u> Practice using rock-like footholds	While bouldering or top roping indoors, restrict footholds to only screw-on holds (aka "jibs"), smears, molded features, or empty bolt holes (divots or ripples on larger holds can also be used).	Toe feel and strain on arms	Steeper terrain, more difficult terrain, faster climbing pace, smaller footholds, polished footholds	Focus on applying as much weight as possible to small footholds, despite any natural reluctance to trust them. Experiment with relaxing the hand's grip until the point of slipping, and then note where that point is.

Advanced Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Glue Feet <u>Objective:</u> Reducing foot slips	On any climbing terrain, climb as though your toes become "frozen" to the wall as soon as they land on the foothold (no pivoting, rotating, or sliding allowed). As you move past the hold, flex at the ankle and forefoot to avoid pivoting on the hold.	Have a partner observe and correct you. Note the "feel" of a properly "glued" foot	Sloper footholds, longer movements, and traversing moves will all increase the difficulty	Focus on establishing a wide contact area between the shoe and hold, and try to maintain that contact area throughout the hold's use. Doing so maximizes holding power and prevents slips.

Dynamic Movement Drills

Beginner Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Two Points of Contact	While climbing on any terrain, remove one foot from the wall (let it dangle) before and during every hand movement. This will ensure you keep only two points of contact (at most) with the rock, and will force a more-dynamic climbing style. It is permissible to have both feet on the wall, just not while reaching with a hand.	Have a partner "referee" this drill	Performing this drill on more-technical and less-steep routes will be more difficult	Focus on selecting footholds that allow smooth, flowing movement. Note how various foothold locations change the body's motion during hand reaches.
One-Arm Traverse	While boulder-traversing on not-so-steep terrain with plenty of holds, climb with only one arm. After completing the traverse, try it again with the other arm. Use big holds, as this drill can be very pumpy.	Observe the effects of dynoing to various hold locations relative to the feet	Steeper terrain, longer reaches, combine with other drills	Focus on involving your entire body in the dyno, not just stabbing your hand to the next hold while your COM stands still. Your body should be moving toward the next static position as the hand is moving.

Intermediate Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Blind Dynos	While bouldering, select a group of handholds that allow you to repeatedly dyno back and forth without moving your feet. Practice these dynos several times, then try it with your eyes closed, using proprioceptive awareness to guide your hands to the hold.	Latching the dynos; note how well you remember the holds' locations and paint a picture in your mind	Increase distance and steepness. Dynos that require hip and COM movement are much harder. Combine with other dyno drills	Focus on remaining relaxed throughout the movement (which can be somewhat frightening). This drill improves your ability to grab holds without deliberate guidance, enabling faster climbing.
Pocket Dynos	On a gym or home wall, set some simple boulder problems that require dynos to tight pockets (may be one-, two-, three-, or four-finger pockets). Rehearse the dynos several times in a session, and perform the drill frequently to develop accuracy.	Catching the hold	Increase the length of the dyno and/or wall steepness. Longer dynos that require the hips and COM to move are much more difficult	Focus on deliberately guiding your fingers into the pocket opening, and hitting it at the deadpoint. Observe the contributions from the other limbs in controlling your movement into the pocket.

Advanced Drills

Name	Setting & Description	Feedback	Progression	Focus on...
Target Campus	On a campus board, use masking tape (or a tick mark) to create "targets" on specified rungs (place the tape on either side of the aim point to create a "virtual pocket"). While campusing, try to hit the targets.	Hitting the target; have a "referee" ensure you don't look away before hitting the target	Reduce the target size, and/or campus farther. Replace the tape boundaries with a dowel or pencil to raise the stakes	Focus on deliberately guiding your fingers to the target and watch your hand land on the hold. This drill will require more body control than a typical campus workout, and can be quite difficult as the gaps increase.
Deadpoints	On a gym or home wall, set boulder-problem dynos with non-positive target holds (slopers, pinches). Latching these holds will require precise deadpoints. Rehearse the dynos repeatedly. An entire route or traverse can be constructed with such moves.	The climber and a partner can observe the overshoot, and how far the COM falls before settling on the hold	Increase distance and steepness, and use less-positive target holds. Dynos that require hip and COM movement are much harder.	Focus on initiating the dyno from the lower body while employing the minimum amount of force to not overshoot (such lower-body control is difficult to execute). Use the trailing hand to control the flight.

SUMMARY

- Rock climbing is among the most technical, skill-dependent sports, yet as beginners, climbers often receive little and/or poor instruction in climbing movement technique. As a result climbers often develop bad habits, requiring extra effort to overwrite them with good habits.
- Apply the *Good Habits for Efficient Skill Development* to daily training and practice sessions. Deliberate practice is the best way to acquire new skills and perfect existing ones.
- Climbers should strive to maximize their time spent *moving over rock*, which often means spending more time on less-difficult terrain, whether training or on goal routes, indoors or outdoors.
- Modern climbing gyms are a wonderful development, but certain of their characteristics encourage sloppy footwork. Use the recommended footwork policies and practice drills to overcome these biases.
- Handwork is an essential skill to getting the most out of a climber's fitness. Experiment with each grip position to find the best fit for a given hold.
- Dynamic movement often makes climbing more efficient, and is a must on some crux moves. The campus board is a great tool for overcoming inhibitions to dynoing.
- Resting is another oft-overlooked skill that is critical to route-climbing success. Be creative, and practice novel rest techniques well before they are needed on redpoint.
- Just like physical training, skill development should be progressive. Several aspects of movement practice can be adjusted to increase the relative difficulty of a particular drill. Use these "controls" to progress the practice toward realistic climbing scenarios.



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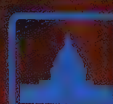
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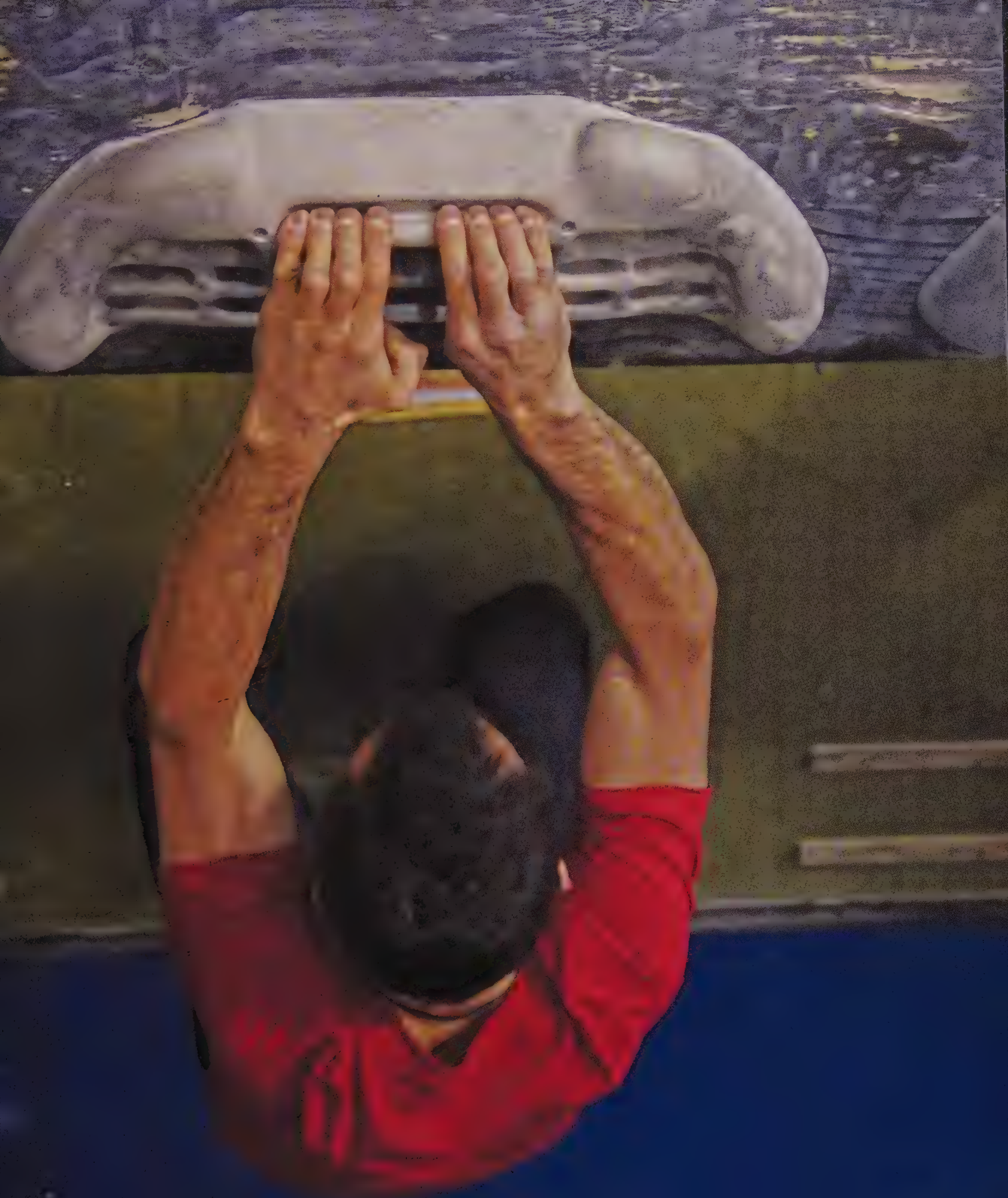
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